

21/07/26

EXP. NO: 1 AZURE DEVOPS ENVIRONMENT SETUP

Aim:

To setup and access the Azure DevOps environment by creating an organization through the Azure Portal.

Installation:

1. Open your web browser and go to the Azure website:

<https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

If you don't have a Microsoft account, you can create one here:

<https://signup.live.com/?lic=1>

Get started with Azure

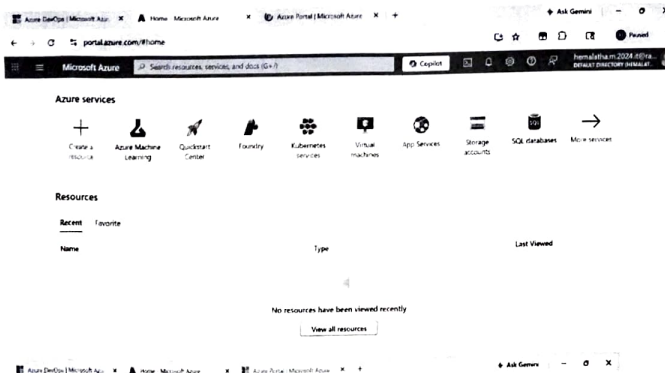
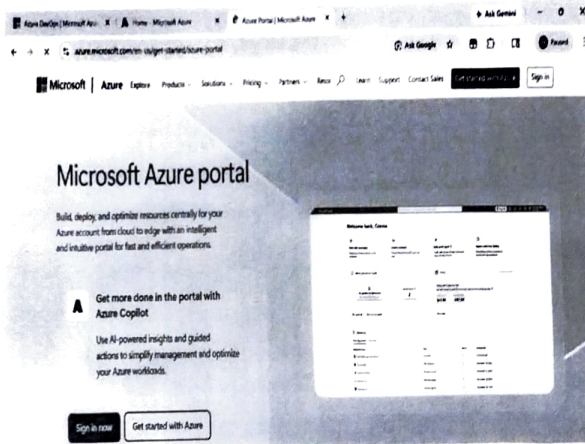
Sign in

2. Azure Home Page

Azure Services - Azure DevOps Organizations

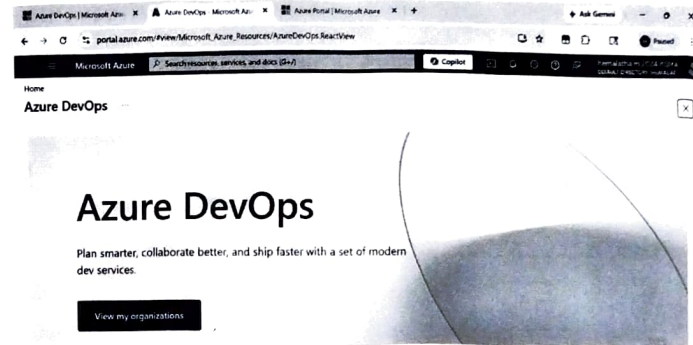
3. Open DevOps environment in the Azure platform by typing Azure DevOps Organizations in the Search bar.

4. Click on the My Azure DevOps Organization link and create an organization and you should be taken to the Azure DevOps Organization Home Page.



Azure DevOps

My Azure DevOps Organizations



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure Portal.

28/11/26

EXP.NO.2 AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient Collaboration and agile work management.

Procedure:

1. Create an Azure Account

Azure DevOps

~~hemalatha.m.2024.it@kajalakshmi.edu.in~~

Name your Azure DevOps Organization

dev.azure.com/hemalatham2024.it

2. Create your first project in your organization

a) After the organization is Set up, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more.

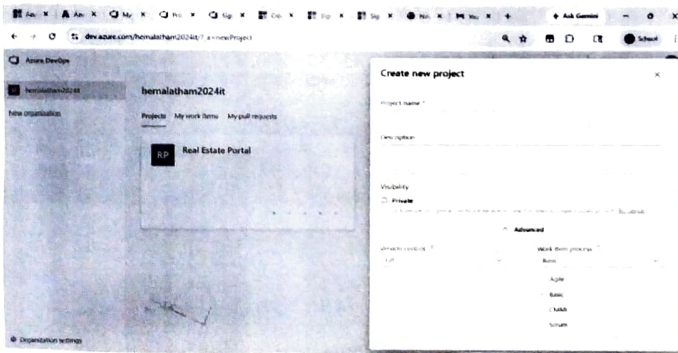
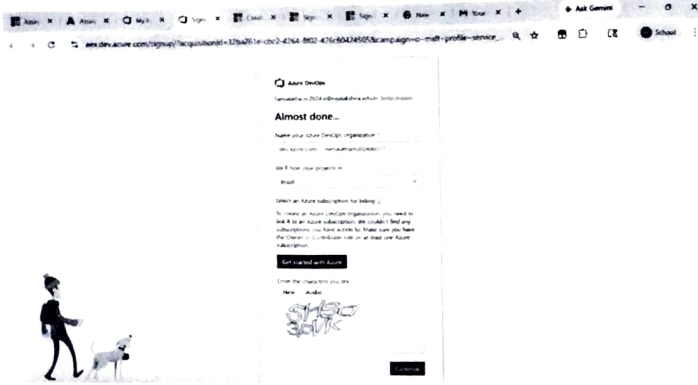
b) On the organization's Home Page, click on the New Project button.

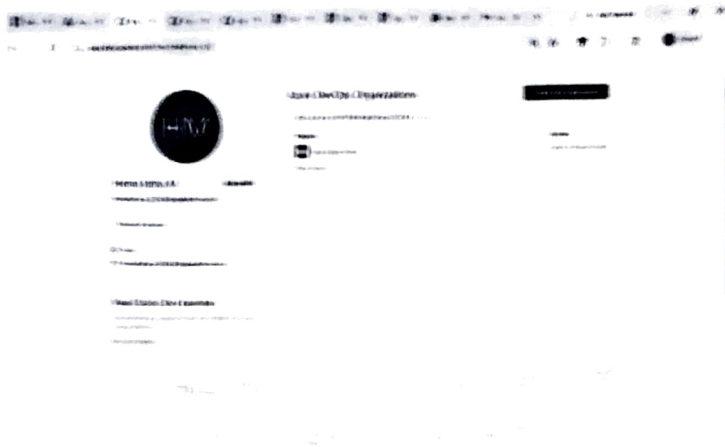
c) Enter the Project name, description and visibility Options:

Name: Choose a name for the project.

Description: Optionally, add a description to provide more context about the project.

Visibility: Choose whether you want the project to be private (accessible only to those visited) or public (accessibilities to anyone)





d) Once you've filled out the details, click
Create to see up your first project

Create New Project

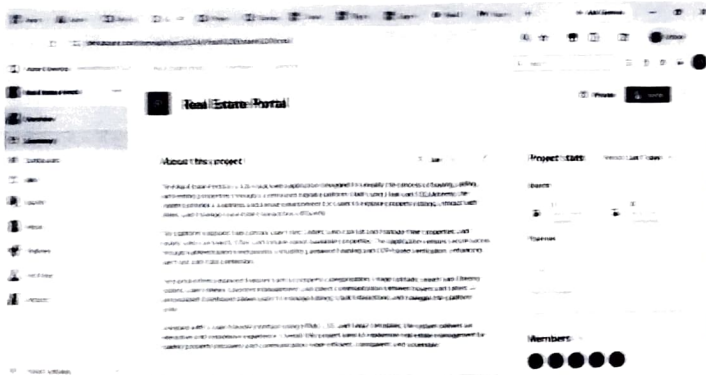
Project Name: Real Estate Portal

Visibility: Public

Version Control: Git

Work Item Process: hema1atha.m.2024.it@rajalakshmi.edu.in

Create



3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the organization name, and you should be taken to the Azure DevOps Organization Home Page.

HM

Hema1atha M

hema1atha.m.2024.it@rajalakshmi.edu.in

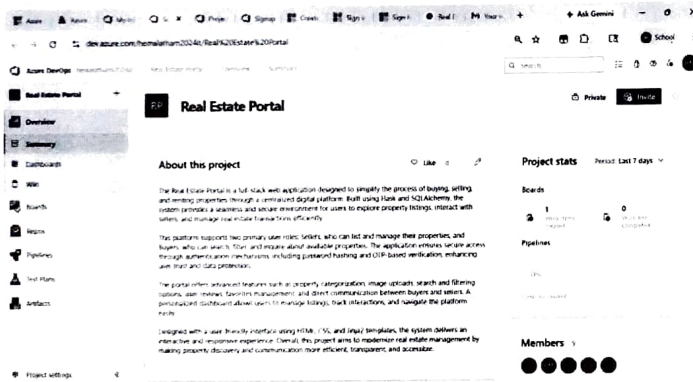
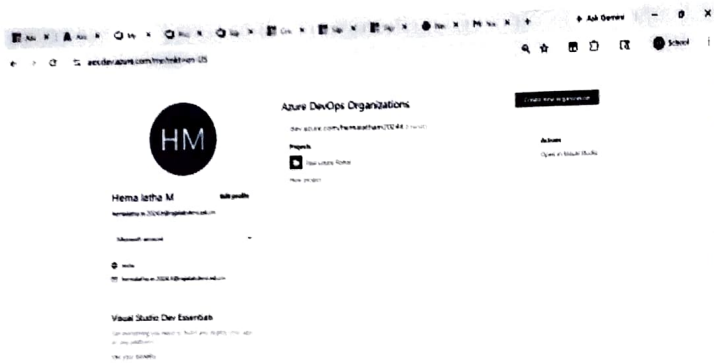
dev.azure.com/hema1atha.m.2024.it (Owner)

Projects - Real Estate Portal

4. Project Dashboard

About this Project

The Advanced Reliable Real Estate Portal is a web application that allows users to list, search and manage properties. It includes features like Secure login,



d) Once you've filled out the details, click
Create to set up your first project.

Create New Project

Project Name: Real Estate Portal

Visibility: Public

Version control: Git

Work item Process: hemalatha.m.2024,it@rajalakshmi.edu.in

Create

3. Once logged in, ensure you are in the correct Organization. If you're part of multiple Organizations, you can switch between them from the top left corner (next to your user profile). Click on the organization name, and you should be taken to the Azure devops Organization Home Page.

(HM)

Hema latha M

hemalatha.m.2024.it@rajalakshmi.edu.in

dev.azure.com/hemalatha.m.2024it (owner)

Projects - Real Estate Portal

4. Project Dashboard

About this Project

The Advanced Reliable Real Estate Portal is a web application that allows user to list, search and manage properties. It includes features like Secure login,

property posting with images, search and filtering, reviews, favourites, and contact seller options. The System provides a user-friendly platform for buyers and sellers to interact efficiently.

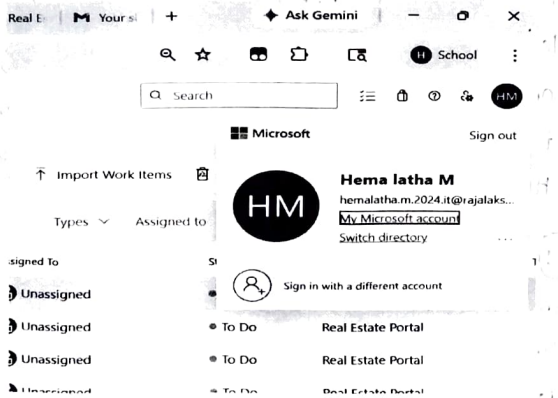
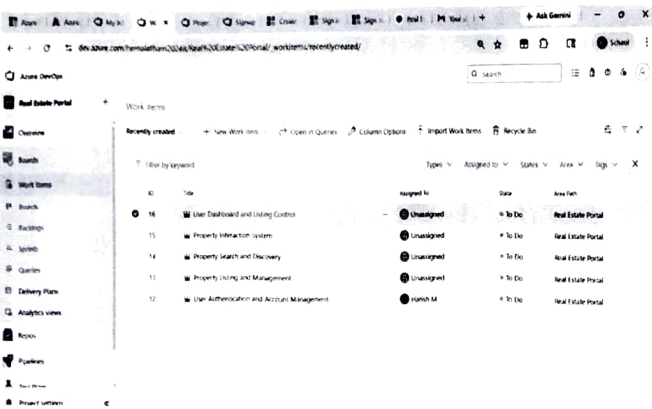
5. To manage user Stories

- From the left-hand navigation menu, click on Boards. This will take you to the main Boards page, where you can manage work items, backlogs and sprints.
- On the work items page, you'll see the option to Add a work item at the top. Alternatively, you can find a + button or Add new work item depending on the view you're in. From the add a work item dropdown, select user story. This will open a form to enter details for the new user story.

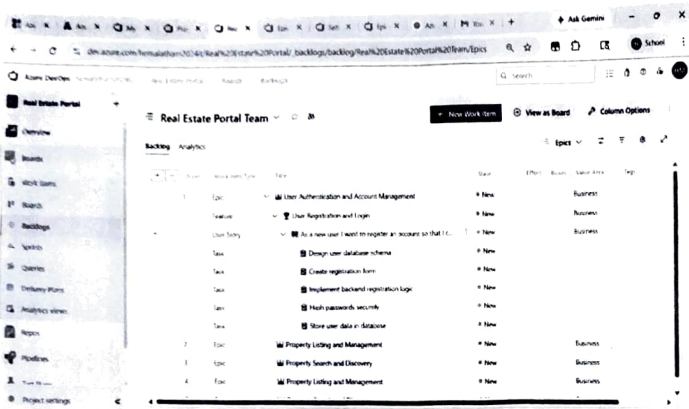
Boards → work items → + New Work Items → user Story

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.



11/2/26



Aim

To learn about how to create epics, user story features, backlogs for your assigned project.

Procedure

Create Epic, Features, User Stories, Task

As a seller, I want to add property details so that buyers can view my property.

1. Fill in Epics

Epic 1

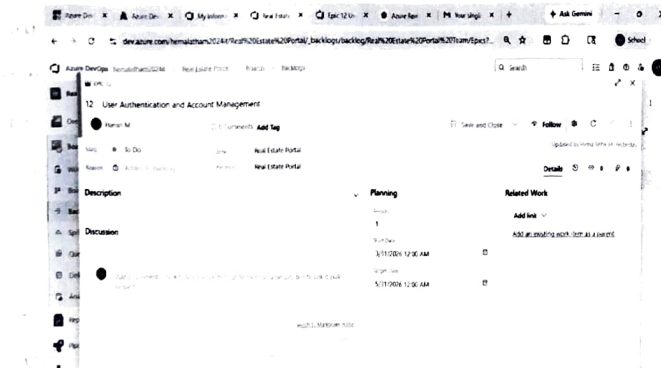
Property Listing Management

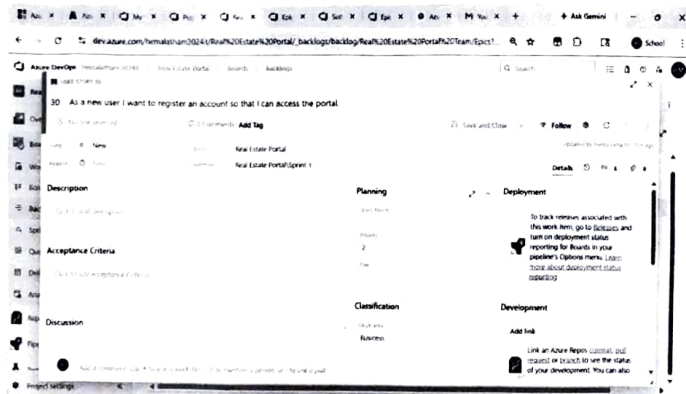
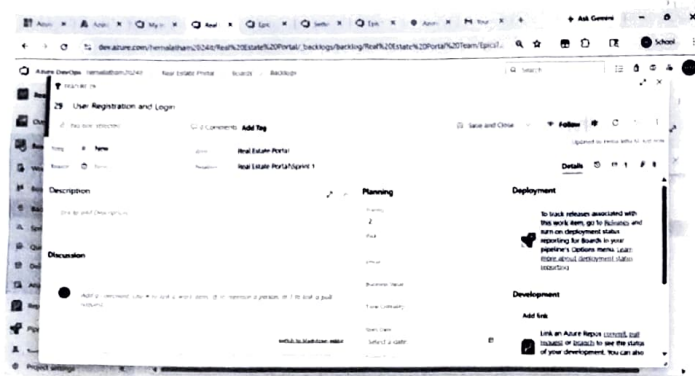
2. Features

- Property Listing Creation
- Property Image Upload
- Property Category Selection
- Property Details Management
- Property Edit and Update
- Property Delete Function

3. Fill in User Story Details

- Design property database schema
- Create property listing form (vs)
- Implement backend logic to save property
- Add image upload functionality
- Validate inputs fields (Price, location, etc..)





13. Implement edit and delete options

14. Test property listing functionality

8. As a user, I want to register and login securely so that I can access the portal.

9. As a seller, I want to upload property images so that listings look attractive.

10. As a buyer, I want to search properties by location and category so that I can find suitable options.

11. As a user, I want to view detailed property information so that I can make decisions.

12. As a user, I want to add properties to favourites so that I can view them later.

13. As a buyer, I want to contact the seller so that I can inquire about the property.

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

25/2/26

EXP.NO. 4

SPRINT PLANNING

Aim:

To assign user story to specific sprint for the Real Estate Portal app.

Sprint planningSprint 1:

Task Board

1. User Registration

Assigned to: Hemalatha M

2. User Login

Assigned to: Hemalatha M

3. OTP verification

Assigned to: Harish Akshay H

4. Password Recovery

Assigned to: Harish DJ

Sprint 2:

Task Board

5. Add property Listing

Assigned to: Harish Akshay H

6. Upload property Images

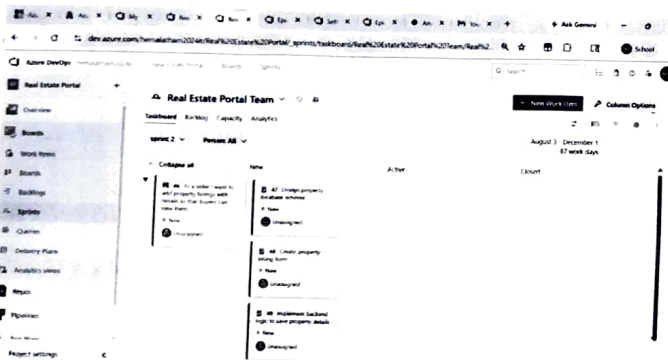
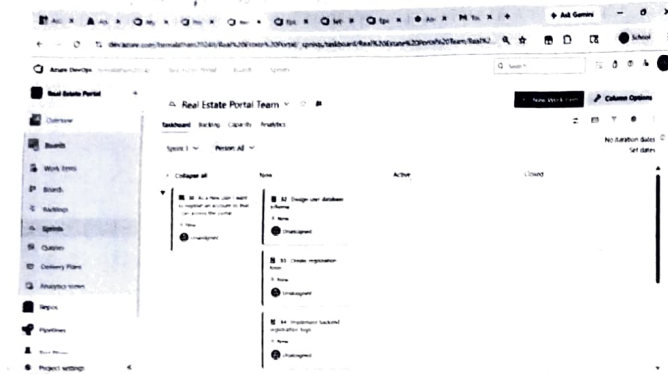
Assigned to: Janani R

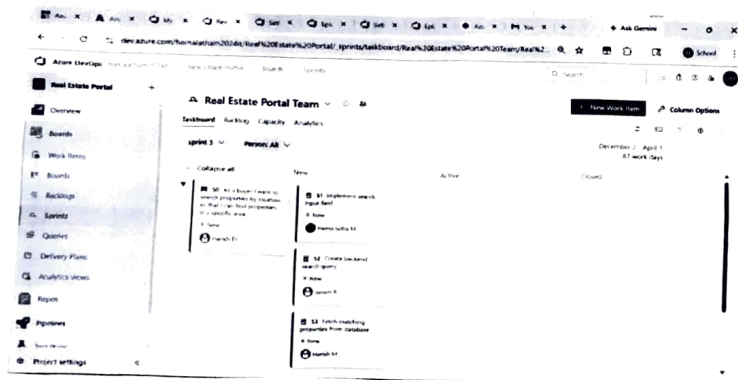
7. Edit property Details

Assigned to: Harish M

8. Delete property

Assigned to: Harish M





Sprint 3:

Task Board

9. Search Property by location

Assigned to: Janani R

10. Filter property by category

Assigned to: Janani R

11. View property details

Assigned to: Harish DS

12. Reviews and ratings

Assigned to: Harish DS

13. Contact seller

Assigned to: Harish Akshay H

14. User dashboard

Assigned to: Hemalatha M

Result:

The sprints were created and tasks were assigned successfully for the Real Estate Portal project.

11/3/26

EXP. NO. 5

POKER ESTIMATION

Aim:

To Create Poker Estimation for the user stories -
Real Estate Portal

Poker Estimation

Story points measure:

- Effort
- Complexity
- Time

This uses Fibonacci scale: 1, 2, 3, 5, 8, 13,

Story Points for Real Estate Portal

User Story	Story Points	Reason
User Registration & Login	5	Authentication + Validation
OTP verification	3	Email + verification Logic
Property Listing	8	Form + database + multiple fields
Image upload	5	File handling + Storage
Property Search	5	Query + filtering
Property Details View	2	Simple data display
Reviews & Ratings	5	User input + Storage

Total = 33 story points

Boards → Work Items → User story → Story Points

↓
migrate (1/2/3/5)

Save

Repeat for all user stories

Process:

Team discussed each user story → compared
Complexity → Assigned story points using Fibonacci
State.

Result:

The Estimation / story points is created for the project
Real Estate portal using poker Estimation.

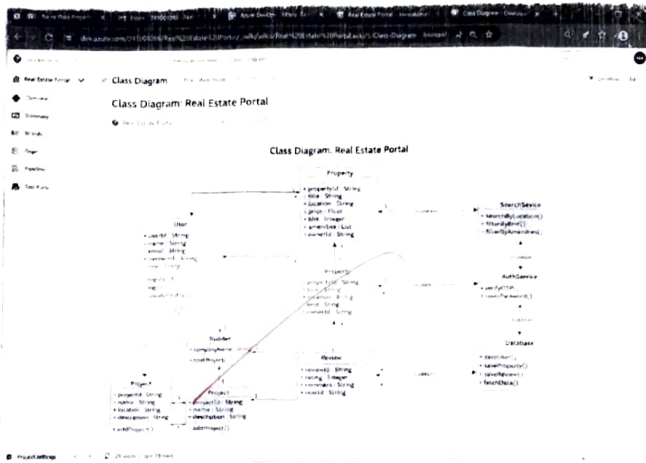
Aim :

6A: Class Diagram

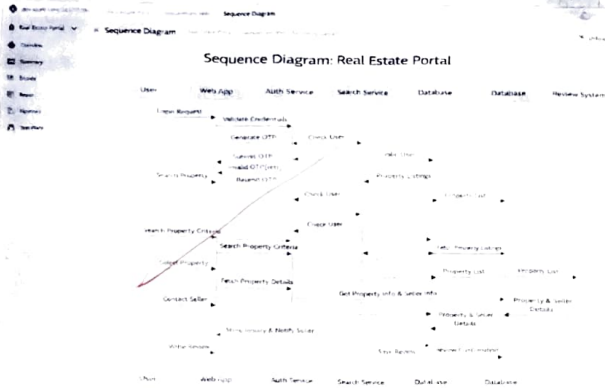
A class diagram is a type of UML diagram that represents the structure of a system by showing its classes, attributes, methods, and relationships between them.

It helps to understand:

- What objects exist in the system
- What ~~data~~ data they store (attributes)
- What actions they perform (methods)
- How they are connected (associations, inheritance, etc)



Sequence Diagram:



6B. Sequence Diagram

Definition:

A Sequence Diagram is a type of UML diagram that shows the interaction between different components of a system over time.

Features:

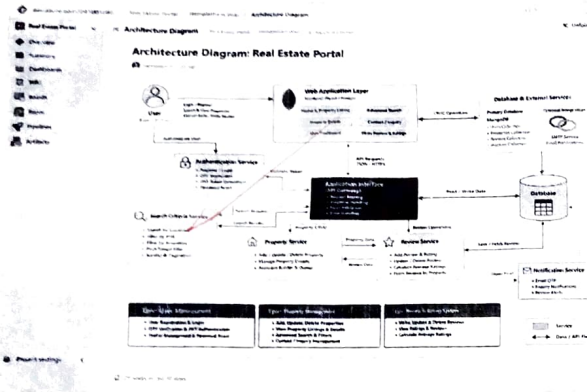
It represents:

- The sequence of messages exchanged
- The order in which operations occur
- The flow of control from one object to another

Result:

The class Diagram and Sequence Diagram is designed successfully for the "Real Estate Portal" project.

Architectural Diagram:



EXP.NO.7 Designing Architectural and ER Diagrams for Project Structure

Aim:

To Design an Architectural Diagram and ER Diagram for the given Project, "Real Estate Portal".

7A. Architectural Diagram

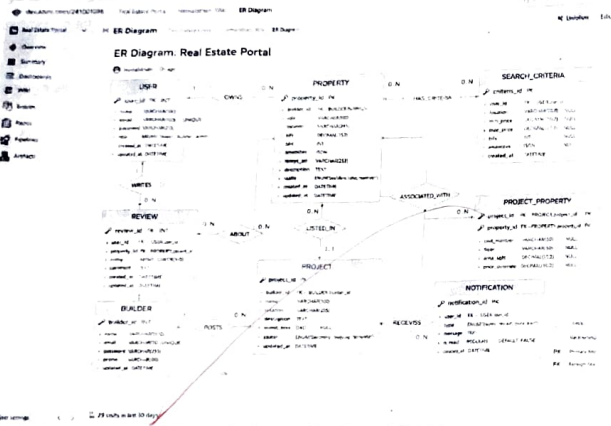
Definition:

Architectural Diagram shows the high-level structure of a system.

Features:

- Illustrates system components and their interactions.
- Shows data/process flows between components.
- Helps to visualize overall architecture of the system.

ER Diagram



7B. ER Diagram

Definition:

An ER Diagram is a graphical representation of a database that shows entities, their attributes, and the relationship between them.

Features:

- Identifies key entities
- Shows attributes of each entity
- Defines relationships between entities (one-to-one, one-to-many, etc.)

Result:

The Architecture Diagram and ER Diagram is designed successfully for the "Real Estate Portal" project.

Aim:

Test plans and test case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test planning and Test case

Test case Design Procedure

1. Understand Core Features of the Application

- User sign & login
- Property search and filtering
- Property Listing (Builder/Admin)
- Viewing Property Details
- Reviews and Ratings
- Property Recommendation system

2. Define User Interactions

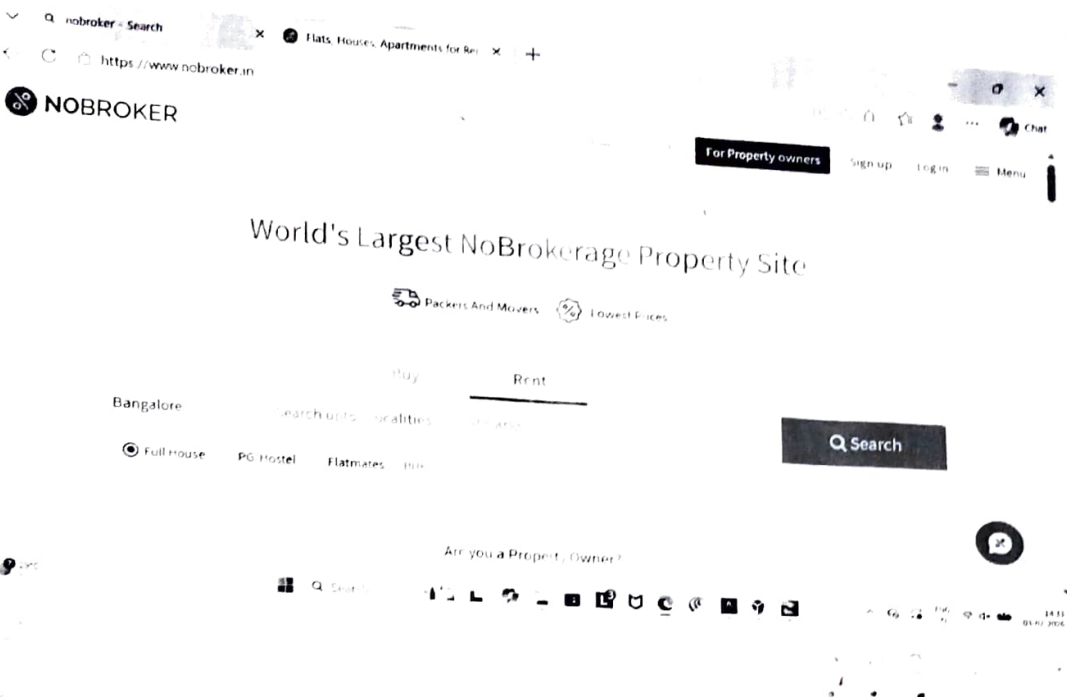
- Each test case represents real user actions.
- Example:
logging in, searching property, adding property, giving review

3. Design Happy path Test cases

- Ensure system works correctly under normal conditions.
- Example: User logs in, searches property, views details successfully.

4. Design Error path Test cases

- Handle invalid or unexpected scenarios
- Example: Wrong login credentials, no search results, failed upload.



Aim:

Test plans and test case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test planning and Test caseTest case Design Procedure1. Understand Core Features of the Application

- User Sign & Login
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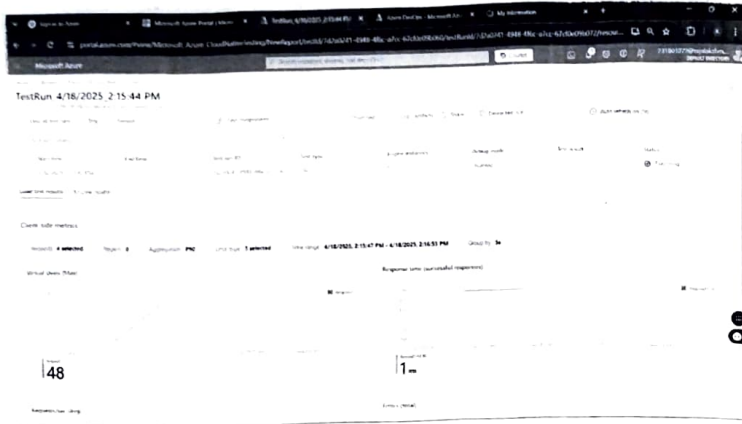
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- Ensure System works correctly under normal Conditions.
- Example: User logs in, searches property, views details successfully.

4. Design Error path Test cases

- Handle invalid or unexpected scenarios
- Example: Wrong login credentials, no search results failed upload.





5. Break Down Steps and Expected Results

- Each test case includes:
 - Actions (steps)
 - Expected outcomes

6. User Clean Naming and IDs

- Example:
 - TC01 - Successful Login
 - TC08 - Property Search Failure

7. Separate Test Suites

- Grouped based on modules
 - Login
 - Property Search
 - Property Management
 - Reviews
 - Recommendations

8. Prioritize and Review

- Important features marked high priority.
- Ensures full coverage of user requirements.

3. TEST CASE

Give two test cases for atleast five user stories showcasing the happy path and error scenarios in Azure DevOps platform.

Real Estate Portal - Test Plans

USER STORIES

1. As a user, I want to sign up and log in securely so that I can access the real estate platform (ID:101).
2. As a user, I want to search and view available properties so that I can find houses that match my preferences (ID:102)
3. As a builder, I want to add and manage property listings so that users can view my properties (ID:103)
4. As a user, I want to view detailed information about a property so that I can make better decisions. (ID:104)
5. As a user, I want to receive property recommendations based on my preferences. (ID:105)

TEST SUITES

Test suit: TSO1 - user Authentication (ID:201)

TC01 - Successful sign up

Action:

- Go to the sign-up page
- Enter valid name, email, and password.
- click "sign up"

Expected Results:

- Sign up form is displayed
- Fields accept values without error
- Account is created successfully and the user is redirected to the dashboard.

Type: Happy Path

TC02 - Secure Login

Action:

- Go to the Login Page
- Enter valid email and password.
- click "Login"

Expected Results:

- Login form is displayed.
- Fields accept data without error
- User is successfully logged in and redirected to the dashboard.

Type: Happy path

TC03 - Sign Up with Existing Email

Action:

- Go to the Sign-up page
- Enter valid name and an already registered email.
- Click "sign up"

Expected Results:

- Fields accept data.
- Error message "Email already registered" is displayed

Type: Error Path

TC04 - Login with Wrong Password

Action:

- Go to the Login Page
- Enter valid email and incorrect password.
- Click "Login"

Expected Results:

- Input is accepted.
- Error message "Invalid email or password" is displayed.

Type: Error path

TEST SUIT : TS02 - PROPERTY SEARCH (ID:202)

TC05 - View Property Listings

Action:

- Login successfully.
- Navigate to the Property Listings section.

Expected Results:

- Available Properties are displayed clearly with images, price and location.

Type: Happy path

TC06 - Property Search Failure

Action:

- Enter an invalid location or filter criteria
- Click search.

Expected Results:

- System Processes the request

- Message "No properties found for the selected criteria" is displayed.

Type: Error path

TEST SUIT: TSD3 - PROPERTY LISTING MANAGEMENT (ID: 203)

TC07 - Add Property Successfully

Action:

- Login as Builder/Admin
- Navigate to Add Property page.
- Enter valid property details.
- click Submit.

Expected Results:

- property details are accepted.
- New Property listing is successfully added to the platform.

Type: Happy Path

TC08 - Property Upload Failure

Action:

- Login as Builder.
- Go to Add Property Page
- Leave required fields empty
- Click submit.

Expected Results:

- System detects missing data.

- Error message " please fill all required fields" is displayed.

Type: Error path

TEST SUIT: TS04 - PROPERTY DETAILS (ID:204)

TC09 - View Property Details

Action:

- Log in to the system
- Select a property from the listing.
- Click View Details.

Expected Results:

- Complete Property information is displayed including price, location, amenities, and images.

Type: Happy path

TC10: Property Details Not Loaded

Action:

- Disconnect from internet or simulate server error.
- Click view Property Details.

Expected Results:

- System fails to retrieve property data
- Error message " unable to load property details" is displayed.

Type: Error path

TEST SUIT : TS05 - PROPERTY RECOMMENDATION SYSTEM (ID:205)

TC11 - Generate Property Recommendations

Actions:

- Login with Valid credentials.
- Select property preferences (location, budget, type).
- click Get Recommendations.

Expected Results:

- System generates recommended properties based on selected filters.

Type: Happy path

TC12 - Recommendation Failure

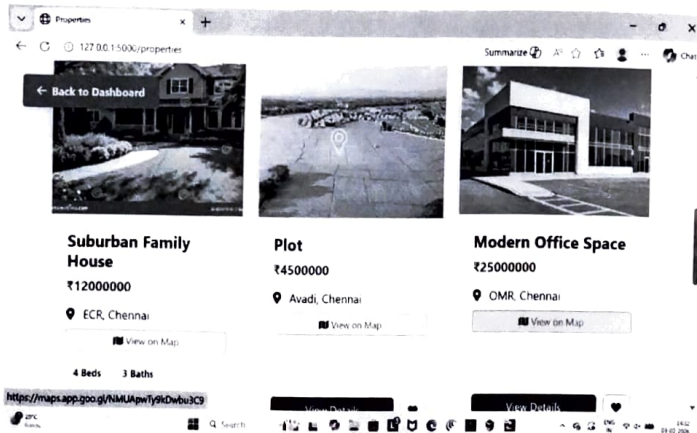
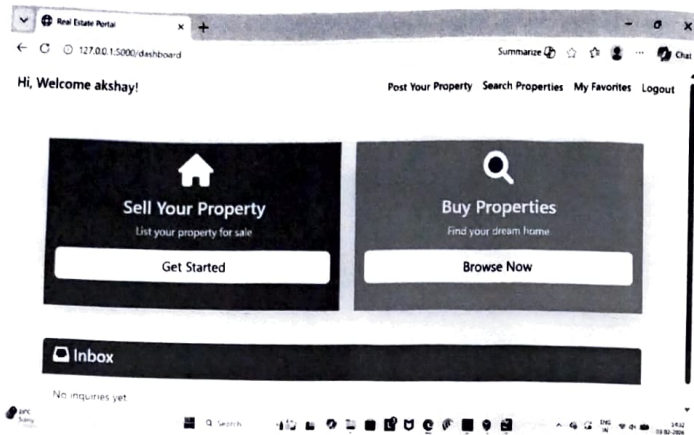
Action:

- Login Successfully.
- Select uncommon or invalid filter combinations.
- click Get Recommendations.

Expected Results:

- System processes request.
- Message "No recommendations found for the selected preferences" is displayed.

Type: Error path



Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests and deployment.

Load Testing:

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure portal

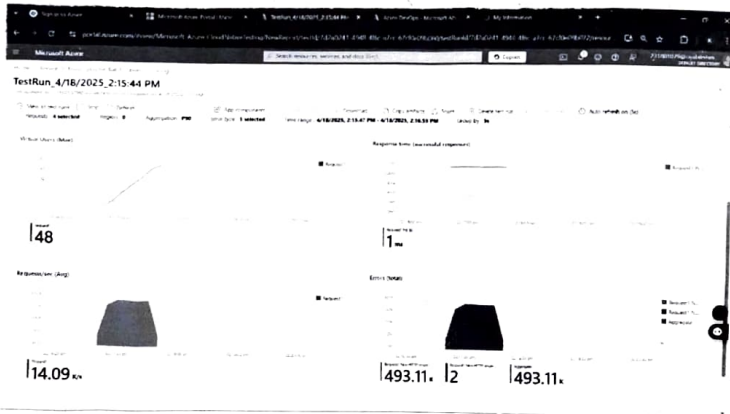
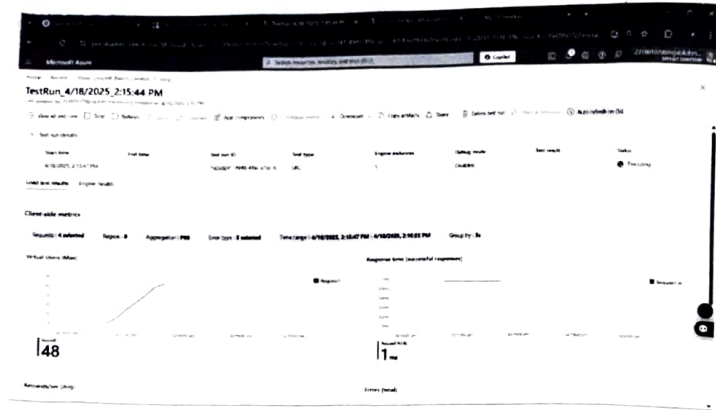
Go to <https://portal.azure.com> and login.

2. Create the resource

- Go to Create the Resource → Search for "Azure Load Testing"
- Select Azure Load Testing and click create.

3. Fill in the Configuration Details

- Subscription: choose your Azure subscription
- Resource Group: Create new or select an existing one.
- Name: provide a unique name (no special characters)
- Location: choose the region for hosting the resource



4. (Optional) Configure tags for categorization and billing

5. Click Review + Create then Create.

6. Once deployment is complete click Go to Resource.

Steps to Create and Run a Load Test

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP Requests > Create.

2. Basic Tab:

- Test Name: Provide a unique Name
- Description: (Optional) Add test purpose
- Run After Creation: keep checked.

3. Load Settings

- Test URL: Enter the target endpoint (e.g. <https://yourapi.com/products>)

4. Click Review + create → create to start the test.

PIPELINES

This experiment demonstrates connecting the Flow-based Real Estate Portal with Azure DevOps. The pipeline installs dependencies, runs basic checks, and ensures booth deployment.

Steps:

Connect Github to Azure DevOps

- Create a new project in Azure DevOps.
- Create a pipeline
- Select Github as source
- Authorize repository access

Create azure-pipelines.yml in Repository Root:

YAML CODE

trigger:

- main

pool:

vmImage: ubuntu-latest

steps:

- checkout: self

- task: use PythonVersion@0

inputs:

versionSpec: '3.x'

displayName: "Set up Python"

- script: |

python -m pip install --upgrade pip

pip install -r requirements.txt

displayName: "Install dependencies"

- script: |

python -c "print('Real Estate Portal pipeline Running successfully')"

displayName: "Run test script"

Pipeline Tasks Include

- Setting up Python Environment
- Installing dependencies
- Running a test script
- Verifying pipeline execution.

Result:

Azure Load Testing resource was created and load test was executed successfully. Azure DevOps Pipeline was created and verified for automated build and execution of the Real Estate Portal.

